JMB Lighting — DMX Personality Chart

JMB Zone / 4our / 8ight · Firmware 2.x · 2026-10-02 (five-channel set added)

Personalities (per port)

The number is the one the panel, the app and an RDM console all show. The RGBW and RGBWA personalities need a **five-output port module** (the .3 boards: outputs red, green, blue, white, amber on XLR pins 2–6, the white and amber terminals printed CW / WW); on a three-output module they are greyed out and cannot be selected.

#	Name	Footprint	Channels
1	Dim 1ch	1	Level
2	Dim 2ch (16-bit)	2	Level coarse, fine
3	CW/WW 8-bit	2	CW, WW
4	CW/WW 16-bit	4	CWc, CWf, WWc, WWf
5	RGB 8-bit	3	R, G, B
6	RGB 16-bit	6	R, Rf, G, Gf, B, Bf
7	**RGB FX**	10	see below
8	RGBW 8-bit	4	R, G, B, W
9	RGBW 16-bit	8	R, Rf, G, Gf, B, Bf, W, Wf
10	**RGBW FX**	12	see below
11	RGBWA 8-bit	5	R, G, B, W, A
12	RGBWA 16-bit	10	R, Rf, G, Gf, B, Bf, W, Wf, A, Af
13	**RGBWA FX**	14	see below

CW/WW on a five-output module drives the white (cool) and amber (warm) terminals and leaves red, green and blue dark; on a three-output module cool rides the green terminal and warm the red one, as it always has.

Dim personalities patch a port like a classic dimmer channel: the one DMX level scales the port's *set colour* (chosen in the app). With no colour set it acts as white — every output carries the level — so a single-colour or **IR strip** lights straight from its one patched channel, whichever output terminal it's wired to.

Each port's whole footprint must fit inside the universe, so the start address caps at **513 − footprint**: Dim 1ch → 512, Dim 2ch → 511, CW/WW 8-bit → 511, CW/WW 16-bit → 509, RGB 8-bit → 510, RGB 16-bit → 507, RGB FX → 503, RGBW 8-bit → 509, RGBW 16-bit → 505, RGBW FX → 501, RGBWA 8-bit → 508, RGBWA 16-bit → 503, **RGBWA FX → 499**. Changing a port to a larger personality re-clamps its start down if it would overhang. Offset apply mode spaces ports by the widest footprint (14ch → 56ch across a 4-port, 112ch across an 8-port).

FX personalities (RGB FX 10ch, RGBW FX 12ch, RGBWA FX 14ch)

CH	Function	Values
1	Dimmer	0–255 master
2–3	Red coarse + fine	16-bit
4–5	Green coarse + fine	16-bit
6–7	Blue coarse + fine	16-bit
8–9	White coarse + fine	16-bit — RGBW FX and RGBWA FX only
10–11	Amber coarse + fine	16-bit — RGBWA FX only
next	Effect select	ranges below (CH8 on RGB FX, CH10 on RGBW FX, CH12 on RGBWA FX)
next	Effect speed	128 = designed tempo; exponential — 0 ≈ ⅛×, 255 = 8×. Beat effects (Club, Party, Pulse, Strobe, Police): 0 stop (Strobe/Police off) · 1–51 ¼× · 52–102 ½× · 103–153 1× · 154–204 2× · 205–255 4× the tempo
last	Strobe	0 = open · 1–128 fixed rate ~0.8→14 Hz · 129–255 random, mean 1 s→60 ms

When the effect channel selects an effect it replaces the colour channels as the colour source (white then follows the white in the effect's mix and amber stays dark); the strobe shutter chops whatever is playing; the dimmer scales last.

Effect ranges (CH8)

21 effects, each an **11-value band** from 10 up (the last extends to 255):

Range	Effect	Range	Effect
0–9	Off (safe park)	120–130	Sunset
10–20	Candle	131–141	TV Sim
21–31	Fire	142–152	Strobe
32–42	Fireworks	153–163	Police
43–53	Water	164–174	Pulse
54–64	Siren	175–185	Party
65–75	Laser	186–196	Traffic
76–86	Club	197–207	Fluoro
87–97	Lightning	208–218	Paparazzi
98–108	Rainbow	219–229	Explosion
109–119	Aurora	230–255	Clouds

Notes: Siren and Police alternate phase across ports (lightbar feel); Club, Party, Pulse, Strobe and Police run on the rig's beat clock (128 BPM by default; the app's tap tempo puts the downbeat on the music). Party hops hue every 4th beat and renders greens as teal — never black, never strobes; Club keeps to deep blue through crimson; Traffic plays passing-headlight sweeps with independent random timing per port; Water, Pulse and Clouds never go fully dark; effect speed (CH9) does not affect the CH10 strobe rate. The film gags: **Fluoro** is a failing fluorescent tube (rapid failed strikes, then it catches and holds with a faint mains buzz); **Paparazzi** fires irregular hard white pops, independent per port, like a press pit; **Explosion** is a white-hot flash blooming to an orange fireball with an ember-flicker decay; **Clouds** drifts daylight slowly dimmer and cooler as cloud passes the sun, then back.

Rig-wide FX controls (config, not DMX): **FX Tempo** (60–240 BPM, tap-settable from the app or set in panel Setup) sets the beat for the beat effects and the speed of the rest, and **FX Spread** (0–100%) staggers the effect clock across ports. Defaults (128 BPM, 0%) reproduce the behaviour above.

Sources

The rig accepts **wired DMX (5-pin)** and **CRMX wireless** (LumenRadio). One source owns the decode at a time, selected in the Sources hub (panel or app).

Hold-on-loss: if the active source drops, every port holds its last look — nothing ever auto-runs on signal loss; changes made after the loss apply immediately. Runt packets (under 25 slots) are rejected as line noise rather than treated as a new look.

RDM

Root device (sub 0): rack footprint, start address = port 1; SET fans out per the Apply mode (JMB parameter 0x8001: 1=Same 2=Seq 3=Offset). Each port is a sub-device (1–N) with its own start address, personality and identify. E1.37-1 CURVE (dimmer curve) and MODULATION_FREQUENCY (PWM library: 1=Smooth 8 kHz, 2=Silent 20 kHz, 3=Camera 15.6 kHz, 4–6=Custom 1–3) are supported on the root and on each port. On the root a read returns port 1 and a write goes to every port.

JMB parameter	ID	On	Value
White balance red	0x8010	root, ports	0–255, 255 = full (also warm on a three-output module)
White balance green	0x8011	root, ports	0–255 (also cool on a three-output module)
White balance blue	0x8012	root, ports	0–255
White balance preview	0x8013	root, ports	0 / 1 — acts only with no DMX present; off after 5 min
White balance white	0x8014	root, ports	0–255 — output 4 of a five-output module (cool on CW/WW)
White balance amber	0x8015	root, ports	0–255 — output 5 (warm on CW/WW)
Custom 1 / 2 / 3 carrier	0x8020 / 0x8022 / 0x8024	root	Hz, 2000–25000; 0 empties the entry. A port on a custom entry keeps its own CURVE

A value outside its range is refused, never altered. The root also exposes each port's temperature and output current as E1.20 sensors, and personality is settable at the root for sub-0-only consoles.